

One Fluid History Through the Terminal Time

Global Closure from Compatible Mixed Sobolev Rectangles for the Three-Dimensional Navier–Stokes Equations

Thomas Birnie

Abstract

For smooth divergence-free finite-energy data on $\mathbb{R}^3$, the datum-generated whole-terminal theorem supplies every finite adjacent Sobolev rectangle of the one maximal Navier–Stokes history. We prove the global closure of that theorem. Direct differentiation produces the pressure-normalized mixed jet. Different finite parts agree because they are derivatives of the same solution, and their overlap gives its complete spatial row, not an infinite norm sum or a sequence of replacement solutions. The equation then makes the nonlinear pressure-completed interaction integrable at every Sobolev order. The velocity reaches the alleged terminal time in every H^m , and the compatible traces define one endpoint in H^∞ . Adjacent rows carry every higher temporal response to that same time. The differentiated pressure and momentum equations identify the actual endpoint jet. A uniform late-slice local construction overlaps the original history, and uniqueness extends it beyond the alleged maximal time. This proves global smoothness from the datum theorem. The critical $H^{3/2}$ dissipation and $H^{1/2}$ height follow as consequences of the same compatible family.

1 The terminal problem belongs to one history

Fix $\nu > 0$ and a real-valued divergence-free datum

$$u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3).$$

We consider the unforced incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = u_0. \quad (1)$$

The pressure is normalized by decay at spatial infinity. Classical local theory gives a unique smooth solution on a maximal interval $[0, T_*)$, where $0 < T_* \leq \infty$; see [2, 3]. Global regularity is the assertion that this maximal time cannot be finite in the whole-space Millennium formulation [1].

The point of departure is simple but decisive. If $T_* < \infty$, the object approaching T_* is still the velocity field generated by u_0 . Its time derivatives, spatial derivatives, pressure responses, and endpoint traces are not rival evolutions. They are simultaneous readings of one motion. The proof succeeds only if every finite reading reaches the same terminal state.

The ordinary energy calculation already preserves part of that identity. Taking the L^2 inner product of (1) with u , using incompressibility, and integrating in time gives

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \, ds = \frac{1}{2} \|u_0\|_2^2 \quad (0 < t < T_*). \quad (2)$$

Thus kinetic energy cannot be the quantity that disappears at a finite endpoint. What remains is a derivative-level problem: can all finite Sobolev readings of the same solution be carried through the time at which classicality was alleged to stop?

The whole-terminal datum theorem answers this question through finite adjacent rectangles. Before giving the definitions, we state the closure it forces.

Theorem 1.1 (Global closure of the datum theorem). *Let $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$ and $\nu > 0$. The unique maximal classical solution of (1) obeying the whole-terminal datum theorem in theorem 3.2 satisfies $T_* = \infty$,*

$$u, p \in C^\infty([0, \infty) \times \mathbb{R}^3),$$

and the energy identity (2) holds for every finite time.

Assume $T_* < \infty$. The energy identity supplies only the base control

$$u \in L^\infty(0, T_*; L^2) \cap L^2(0, T_*; H^1).$$

It does not place a chosen higher response and its time derivative in an adjacent Sobolev pair uniformly through T_* . Such a pair is exactly what creates a middle-order trace. The datum theorem in section 3 supplies every fixed finite pair on the whole terminal interval. Their shared entries are derivatives of the same history, so the complete spatial row reaches one H^∞ endpoint. The equation recovers the actual pressure-complete jet there, and a late-slice restart continues the history past T_* . No bound uniform over all derivative orders is used.

2 The equation generates its pressure-complete jet

Taking the divergence of (1) and using $\nabla \cdot u = 0$ gives

$$-\Delta p = \partial_i \partial_j (u_i u_j), \quad p = R_i R_j (u_i u_j). \quad (3)$$

Here R_i denotes the i -th Riesz transform. The second identity fixes the chosen decaying pressure and makes its dependence on the whole velocity field explicit. Pressure is a nonlocal simultaneous response of the same history, not an additional datum.

Set

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p.$$

Leibniz' rule and (3) give, for every $n \in \mathbb{N}_0$,

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}, \quad (4)$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n. \quad (5)$$

The momentum recurrence advances the velocity response by one time derivative. The pressure recurrence applies the incompressibility correction at precisely the same temporal depth. Omitting either formula breaks the identity of the differentiated history.

Spatial differentiation opens an independent direction. For a multi-index $\alpha \in \mathbb{N}_0^3$, define

$$J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p.$$

Since ∂_t commutes with fixed spatial derivatives, a second use of Leibniz' rule yields

$$\begin{aligned} J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} + \nabla \Pi_{n,\alpha} \\ = \nu \Delta J_{n,\alpha}, \quad \nabla \cdot J_{n,\alpha} = 0. \end{aligned} \quad (6)$$

The temporal index counts response at a fixed spatial point; the multi-index counts spatial resolution of that response. The Laplacian moves two spatial orders, the transport term distributes derivatives between two velocity factors, and pressure couples the result across space. This is why the two indices must remain independent.

At $t = 0$, the same recurrence generates the entire finite initial jet. Write $V_n^0 = V_n(0)$ and $Q_n^0 = Q_n(0)$. Beginning with $V_0^0 = u_0$,

$$\begin{aligned} Q_n^0 &= R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i}^0 V_{n-a,j}^0, \\ V_{n+1}^0 &= \nu \Delta V_n^0 - \sum_{a=0}^n \binom{n}{a} (V_a^0 \cdot \nabla) V_{n-a}^0 - \nabla Q_n^0. \end{aligned}$$

Sobolev multiplication and boundedness of the Riesz transforms preserve every finite Sobolev order. Induction gives

$$V_n^0, Q_n^0 \in H^m(\mathbb{R}^3) \quad \text{for every finite } n, m. \quad (7)$$

The initial condition fixes every finite jet value. What is not yet known is whether the corresponding response rows remain integrable with constants uniform as the terminal cutoff approaches T_* .

3 Whole-terminal adjacent rectangles

3.1 Why the basic unit has two adjacent rows

Suppose a Hilbert-valued function v satisfies

$$v \in L^2(0, T_*; H^{m+1}), \quad v_t \in L^2(0, T_*; H^{m-1}).$$

The first row measures one derivative more than the desired trace; the second measures one derivative less. Their pairing lands exactly at the middle order. For smooth v ,

$$\|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2 = 2 \operatorname{Re} \int_r^t \langle v_t(s), v(s) \rangle_{H^{m-1}, H^{m+1}} \, ds. \quad (8)$$

Cauchy–Schwarz gives

$$\left| \|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2 \right| \leq 2 \|v_t\|_{L^2(r,t; H^{m-1})} \|v\|_{L^2(r,t; H^{m+1})}. \quad (9)$$

Density extends the calculation to the stated spaces. In particular, v has a unique representative in $C([0, T_*]; H^m)$; this is the standard Lions–Magenes trace mechanism [5].

The estimate explains the geometry needed at the terminal time. One isolated Sobolev row does not carry its own time trace. An adjacent pair does. A finite rectangle stores these pairs for finitely many temporal rows and one chosen spatial depth.

Definition 3.1 (Finite mixed rectangle). For $N, M \in \mathbb{N}_0$ and $0 < T < T_*$, set

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) := & \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 \\ & + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2. \end{aligned} \quad (10)$$

For $M = 0$, the lower space is the Bessel-potential space H^{-1} . The object $\mathcal{R}_{N,M}[u_0; T]$ is the displayed velocity block together with the identities (4)–(6) in that finite coordinate range. Its pressure rows are not assigned an independent norm: after a target order m is fixed, any rectangle with spatial depth at least $m + 2$ gives

$$Q_n \in L^1(0, T; H^m), \quad \nabla Q_n \in L^1(0, T; H^{m-1}) \quad (0 \leq n \leq N)$$

by Sobolev multiplication and Riesz boundedness. We write $\mathcal{R}_{N,M}[u_0]$ for the same finite tuple on $I_* = (0, T_*)$ when its whole-terminal norm is finite.

The notation does not introduce a new solution. Every entry of $\mathcal{R}_{N,M}[u_0; T]$ is the corresponding derivative of the maximal history already fixed in [section 2](#).

3.2 The datum-generated terminal estimate

The closure argument begins at the following datum theorem, the first theorem of the one-fluid compatible-intersection program [\[7\]](#). It is the only nonstandard input used below.

Theorem 3.2 (Datum input: whole-terminal compatible rectangles). *Let (u, p) be the maximal history generated by $u_0 \in \mathcal{S}_\sigma(\mathbb{R}^3)$, and suppose $T_* < \infty$. For every finite $N, M \in \mathbb{N}_0$, there is a constant $C_{N,M} = C_{N,M}(u_0, \nu, T_*) < \infty$ such that*

$$\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) \leq C_{N,M}. \quad (11)$$

If $N' \leq N$ and $M' \leq M$, restriction to the lower temporal rows and the continuous Sobolev embeddings give

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0] = \mathcal{R}_{N',M'}[u_0]. \quad (12)$$

Every shared entry is the same derivative of the same pressure-complete history.

The decisive content of [theorem 3.2](#) is its rowwise datum estimate: for every fixed finite (n, m) ,

$$V_n \in L^2(0, T_*; H^{m+1}), \quad V_{n+1} \in L^2(0, T_*; H^{m-1}), \quad (13)$$

with a bound depending on u_0, ν, T_*, n, m and uniform in the cutoff $T < T_*$. This is the datum-to-terminal payment. It is not a consequence of strict-subinterval smoothness or of the finite summation below.

Proposition 3.3 (Finite-coordinate realization). *The rowwise datum estimate (13) is equivalent to the family of finite bounds (11), and the resulting finite blocks obey (12).*

Proof. Monotone convergence in the terminal cutoff turns each terminal-uniform rowwise bound into the corresponding $L^2(0, T_*; \cdot)$ membership. Fix (N, M) . Adding the finitely many squared adjacent-row bounds for $0 \leq n \leq N$ gives (11). Conversely, the collection of (11) over all finite (N, M) recovers every instance of (13). The recurrences (4)–(6) identify all entries with actual derivatives of the one maximal history. Restricting a larger finite block leaves each shared derivative unchanged, which proves (12). $\square$

Three quantifiers in [theorem 3.2](#) carry the argument. The coordinate (N, M) is fixed before its constant is chosen. The same constant then controls every terminal cutoff $T < T_*$. No constant independent of (N, M) occurs. Thus the theorem provides every finite view of one history without asking an infinite sum of Sobolev norms to converge.

The distinction from strict-subinterval regularity can be written in one line:

$$\underbrace{\forall T < T_* \ \mathfrak{R}_{N,M}(u; T) < \infty}_{\text{classicality before the endpoint}} \not\equiv \underbrace{\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty}_{\text{whole-terminal control}}.$$

The theorem supplies the expression on the right. This is what allows the adjacent trace estimate to see T_* .

4 The complete zero row reaches one endpoint

Increasing N reveals more temporal responses and increasing M reveals more spatial resolution. Every overlap is literal equality of derivatives of u . In particular, the zeroth temporal entry of the family gives

$$u \in \mathcal{S}_0(0, T_*) := \bigcap_{m=0}^{\infty} L^2(0, T_*; H^m(\mathbb{R}^3)). \quad (14)$$

For each fixed m , this is one finite membership supplied by a rectangle of sufficient spatial depth. The intersection records all such memberships of the same field; it does not sum their norms.

Write the pressure-completed interaction as

$$\mathcal{B}(u, u) := (u \cdot \nabla)u + \nabla p = \mathbb{P} \nabla \cdot (u \otimes u). \quad (15)$$

The Sobolev product theorem and boundedness of the Leray projection give, for each integer $m \geq 0$,

$$\|\mathcal{B}(u, u)\|_{H^m} \leq C_m \|u \otimes u\|_{H^{m+1}} \leq C_m \|u\|_{H^{m+2}}^2. \quad (16)$$

Standard product and multiplier facts used here are recalled in [section A](#); see also [\[4, 6\]](#).

Assume for contradiction that $T_* < \infty$. From [\(14\)](#) and [\(16\)](#),

$$\int_0^{T_*} \|\mathcal{B}(u, u)(t)\|_{H^m} dt \leq C_m \|u\|_{L^2(0, T_*; H^{m+2})}^2 < \infty. \quad (17)$$

The viscous term obeys

$$\int_0^{T_*} \|\Delta u(t)\|_{H^m} dt \leq \sqrt{T_*} \|u\|_{L^2(0, T_*; H^{m+2})} < \infty. \quad (18)$$

The original equation, now written as

$$u_t = \nu \Delta u - \mathcal{B}(u, u),$$

gives

$$u_t \in L^1(0, T_*; H^m) \quad \text{for every finite } m. \quad (19)$$

Since $u \in L^2(0, T_*; H^m) \subset L^1(0, T_*; H^m)$, we have

$$u \in W^{1,1}(0, T_*; H^m) \hookrightarrow C([0, T_*]; H^m). \quad (20)$$

More explicitly,

$$\begin{aligned} \sup_{0 \leq t \leq T_*} \|u(t)\|_{H^m} &\leq \|u_0\|_{H^m} + \nu \sqrt{T_*} \|u\|_{L^2(0, T_*; H^{m+2})} \\ &\quad + C_m \|u\|_{L^2(0, T_*; H^{m+2})}^2. \end{aligned} \quad (21)$$

Every quantity on the right is finite at its fixed spatial order. If $m_2 > m_1$, the two continuous representatives agree almost everywhere in H^{m_1} , hence everywhere. They define one distribution

$$u_* := u(T_*) \in \bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3). \quad (22)$$

For any sequence $t_j \uparrow T_*$, the same velocity $u(t_j)$ converges to u_* in every fixed H^m .

The other temporal rows reach this same time by their adjacent pairs.

Lemma 4.1 (Compatible traces of every response). *For every finite n, m , the response V_n has a unique representative in*

$$V_n \in C([0, T_*]; H^m(\mathbb{R}^3)). \quad (23)$$

Representatives at different Sobolev orders agree after the natural embedding. For $n = 0$, this representative is the one constructed in (20) and its terminal value is u_ .*

Proof. Choose a rectangle containing

$$V_n \in L^2(0, T_*; H^{m+1}), \quad \partial_t V_n = V_{n+1} \in L^2(0, T_*; H^{m-1}).$$

The adjacent trace estimate (9) gives (23). Two representatives at different orders equal the same distribution almost everywhere. Their difference is continuous in the lower space and vanishes almost everywhere, so it vanishes at every time. The $n = 0$ representative also equals the $W^{1,1}$ representative almost everywhere and hence everywhere. $\square$

5 The endpoint carries the actual pressure-complete jet

An H^∞ velocity endpoint is enough to invoke standard local theory. The stronger statement needed to preserve the differentiated history is also available: every left temporal derivative and its normalized pressure reaches the endpoint, and the limiting values satisfy the same recurrence.

For $n \in \mathbb{N}_0$, set

$$v_n := V_n(T_*).$$

These values exist in every H^m by lemma 4.1. Define the normalized endpoint pressure from the velocity traces by

$$q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} v_{a,i} v_{n-a,j}. \quad (24)$$

The endpoint momentum identity to be verified is

$$v_{n+1} = \nu \Delta v_n - \sum_{a=0}^n \binom{n}{a} (v_a \cdot \nabla) v_{n-a} - \nabla q_n. \quad (25)$$

Proposition 5.1 (Endpoint identification). *For every finite n, m ,*

$$v_n, q_n \in H^m(\mathbb{R}^3),$$

and

$$v_n = \lim_{t \uparrow T_*} \partial_t^n u(t), \quad q_n = \lim_{t \uparrow T_*} \partial_t^n p(t) \quad \text{in } H^m. \quad (26)$$

Thus $(v_n, q_n)_{n \geq 0}$ is the actual left endpoint jet of the maximal history.

Proof. Every v_n belongs to H^∞ by lemma 4.1. Sobolev multiplication and Riesz boundedness place the right side of (24) in every H^m , so $q_n \in H^\infty$. On $(0, T_*)$, the expressions in (4)–(5) equal the actual derivatives Q_n and V_{n+1} . Every velocity factor is continuous to T_* in arbitrarily high Sobolev spaces by lemma 4.1. The product estimates and Riesz boundedness allow passage to the endpoint in each fixed H^m . The pressure limit is (24); the momentum limit equals both v_{n+1} , by continuity of V_{n+1} , and the right side of (25). This proves the endpoint identity and both limits in (26). $\square$

The pressure normalization matters here. Adding a different time-dependent constant to each preterminal pressure would create a formal ambiguity in the scalar pressure values, even though its gradient is unchanged. The decaying Riesz representation (3) fixes one pressure at every time and passes that same normalization through the endpoint recurrence.

6 Restart of the same history

We now close the contradiction at the single Sobolev level H^3 . The endpoint $u_* \in H_\sigma^\infty$ lies in H_σ^3 , and equation (21) gives

$$K_3 := \sup_{0 \leq t \leq T_*} \|u(t)\|_{H^3} < \infty.$$

Standard local well-posedness gives a lifespan $\delta = \delta(\nu, K_3) > 0$ for every datum in the H^3 -ball of radius K_3 . Choose $t_0 < T_*$ so close to the endpoint that $T_* - t_0 < \delta/2$. The local solution launched from $u(t_0)$ exists at least until $t_0 + \delta > T_*$. On $[t_0, T_*)$ it shares its datum with the original maximal history, so strong uniqueness identifies the two solutions there. It is an extension of that history beyond T_* .

The embedding $H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3)$ closes the transport estimate and strong uniqueness in this local theorem. Because the extending solution equals u on the open overlap, all of their velocity derivatives and decaying pressure responses agree there. Continuity to T_* , together with proposition 5.1, identifies the right endpoint jet of the extension with the original left endpoint jet. The continuation is one smooth pressure-complete history, not only an H^3 extension.

This extension contradicts the definition of T_* as the maximal classical time. The assumption $T_* < \infty$ is impossible. We have proved $T_* = \infty$, and Sobolev embedding at each finite time gives

$$u, p \in C^\infty([0, \infty) \times \mathbb{R}^3).$$

The energy identity (2) remains valid on every finite interval. Together with the Schwartz datum, divergence constraint, zero forcing, and decaying pressure normalization fixed at the outset, this supplies the whole-space finite-energy formulation. This completes the proof of theorem 1.1.

7 The critical-height block inside the same family

The global contradiction used the complete family directly. Its familiar critical quantities can now be read without turning one selected height into a producer condition. Put

$$\Lambda = (-\Delta)^{1/2}, \quad E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t).$$

Pairing (1) with $\Lambda^{2s}u$ gives

$$E'_s = -2\nu E_{s+1} + \mathcal{P}_s, \quad (27)$$

where

$$\mathcal{P}_s := -\operatorname{Re} \langle \Lambda^s u, \Lambda^s ((u \cdot \nabla)u + \nabla p) \rangle. \quad (28)$$

The pressure gradient cancels in this scalar pairing, while its whole-field action remains in the velocity–pressure recurrence.

Corollary 7.1 (Adjacent critical readout). *The rectangle family gives*

$$E_{3/2} \in L^\infty(0, T_*) \cap L^1(0, T_*), \quad H \in W^{1,1}(0, T_*), \quad \mathcal{P}_{1/2} \in L^1(0, T_*). \quad (29)$$

Proof. The rectangle at integer spatial depth $M = 2$ contains the stronger rows

$$u \in L^2(0, T_*; H^3), \quad u_t \in L^2(0, T_*; H^1).$$

The trace estimate first gives $u \in C([0, T_*]; H^2)$, hence $u \in C([0, T_*]; H^{3/2})$ and $E_{3/2} \in L^\infty(0, T_*)$. Its L^1 membership follows from $u \in L^2(0, T_*; H^3)$. The same rows give

$$H'(t) = \operatorname{Re} \langle \Lambda^{1/2} u_t(t), \Lambda^{1/2} u(t) \rangle \in L^1(0, T_*),$$

and $H \in L^1(0, T_*)$. Hence $H \in W^{1,1}(0, T_*)$. At $s = 1/2$, (27) reads

$$\mathcal{P}_{1/2} = H' + 2\nu E_{3/2},$$

whose right side lies in $L^1(0, T_*)$. □

A Analytic facts used by the proof

We collect the standard estimates used above, both to fix conventions and to make their finite-order dependence explicit.

A.1 Sobolev multiplication

For $s > 3/2$, $H^s(\mathbb{R}^3)$ is a Banach algebra. More generally, for each integer $m \geq 0$, the Leibniz and Sobolev embedding estimates give

$$\|fg\|_{H^{m+1}} \leq C_m \|f\|_{H^{m+2}} \|g\|_{H^{m+2}}. \quad (30)$$

Applying $\nabla \cdot$ loses one spatial derivative. Since the Leray projection is a matrix of the identity and compositions of Riesz transforms, it is bounded on every H^m . Formula (30) gives (16). These facts are standard; see [4, 6].

A.2 The adjacent Hilbert triple

Let

$$X_1 = H^{m+1}(\mathbb{R}^3), \quad X_0 = H^m(\mathbb{R}^3), \quad X_{-1} = H^{m-1}(\mathbb{R}^3).$$

The dense continuous embeddings $X_1 \hookrightarrow X_0 \hookrightarrow X_{-1}$ identify X_{-1} with the dual of X_1 through the pivot X_0 . If $v \in L^2(0, T; X_1)$ and $v_t \in L^2(0, T; X_{-1})$, smooth temporal convolution gives approximants v_ε for which (8) is classical. The two L^2 convergences pass the right side to the limit, while (9) makes the X_0 -norms Cauchy uniformly at Lebesgue times. The limit is the unique X_0 -continuous representative of v . This is the form of the Lions–Magenes lemma used in lemma 4.1.

A.3 Spatial smoothness from all finite orders

For each integer $k \geq 0$, choose $m > k + 3/2$. The Sobolev embedding

$$H^m(\mathbb{R}^3) \hookrightarrow C_b^k(\mathbb{R}^3)$$

shows

$$\bigcap_{m < \infty} H^m(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

The constants depend on k and m . No uniform constant over all derivative orders is involved.

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